Expt : 4

18/2/26

Interfacing DAC

Aim :

To write an assembly language program in 8086 to generate squtooth and square waveforms using DAC.

Apparatus Required

S.no	Item	Specification	Quantity
1.	Microprocessor kit	8086	1
2.	Power supply	5V, DC, 12VDC	1
3.	DAC interterance board	—	1

Program : Square wave

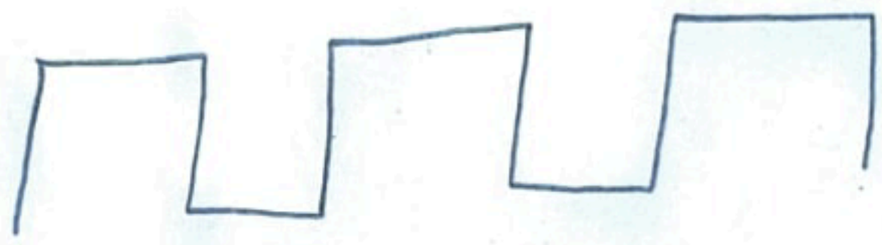
Memory location	Program	comments
1000	MOV AL, 00H	Load 00H in accumulator
1003	OUT 0C8H-AL	Send through output port
1005	CALL DELAY(1100)	call program to 1100
1008	MOV AL, 0FFH	Load 00H in accumulator
100A	OUT 0C8H, AL	Send through output port
100D	CALL DELAY(1100)	call program to 1100
1100	MOV CX, 0505	Load 0505H in accumulator
1103	DEC CX	Decrement CX
1105	JNZ 1104	Repeat until zero
1108	RET	Return to main program

TABULATION :

Waveform	Amplitude	Time Period
Sautooth	5V	1.48 ms
Square	5V	9.8ms

MODEL GRAPH

Square wave :



Sautooth wave :

Program : Sauthooth wave

Memory location	Program	comments
1000	START: MOV AL, 00H	Load 00H in accumulator
1003	LOOP : OUT 0C8H, AL	Send through output port
1005	INC AL	Increment contents of accumulator
1007	JNC LOOP	Jump if no carry
1009	JMP START	Go to starting location

Result:

Thus, the sautooth and square wave form has been generated.